QUALIFICATION
REQUIREMENTS
REMOVED

MIL-M-38510/33A
8 August 1983
SUPERSEDING
MIL-M-38510/33
30 June 1975

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, DTL, FLIP FLOPS, MONOLITHIC SILICON

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS REVISION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, DTL, logic microcircuits. One product assurance class and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The complete part number shall be in accordance with MIL-M-38510, with the exception that the "JAN" or "J" certification shall not be used.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	RS/JK clocked flip flop
02	RS/JK clocked flip flop
03	Pulse triggered binary
04	Dual JK flip flop

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4"), flat package
B	F-3 (14-lead, 3/16" x 1/4"), flat package
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +8.0 V dc
Input voltage range	-1.5 V dc to +5.5 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation, per gate (P_D)	8.5 mW dc
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Cases A, B, and D	0.09°C/mW
Case C	0.08°C/mW
Junction temperature (T_J)	+175°C

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage range (V_{CC})	+4.5 V dc minimum to +5.5 V dc maximum
Minimum high-level input voltage (V_{IH})	+1.9 V dc @ +25°C
Maximum low-level input voltage (V_{IL})	+1.1 V dc @ +25°C
Normalized fanout (each output):	
Device types 01 and 04	10 maximum
Device type 02	9 maximum
Device type 03	8 mmaximum
Case operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Logic diagram and terminal connections. The logic diagram and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables. The truth tables shall be as specified on figure 2.

3.2.3 Case outlines. Case outlines shall be as specified in MIL-M-38510 and in 1.2.3 herein.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2 herein. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container. The "CAN" or "J" certification mark shall not be used.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/	Device type	Limits Min	Limits Max	Unit
High level output voltage	V_{OH}	V_{CC} = 4.5 V	01,02,04	3.3		Volt
			03	2.5		
Low level output voltage	V_{OL}	V_{CC} = 4.5 V	All		0.45	Volt
High level input current	I_{IH1}	V_{CC} = 5.5 V	01,02,04		5.0	µA
High level input current at clear and set	I_{IH2}	V_{CC} = 5.5 V	01,02,04		10	µA
			03		5	
High level input current at clock	I_{IH3}	V_{CC} = 4.0 V	01,02,04		20	µA
			03		30	
Low level input current	I_{IL1}	V_{CC} = 5.5 V	01,02		-1.07	mA
			03		-2.4	
			04		-0.98	
Low level input current at clear, set, and clock	I_{IL2}	V_{CC} = 5.5 V	01,02		-3.2	mA
			03		-1.82	
			04		-2.9	
Short circuit output current	I_{OS}	V_{CC} = 5.5 V, V_{IN} = GND	01	-.615	-1.34	mA
			02	-1.84	-4.02	
			03	-14	-29	
			04	-.62	-1.33	
Supply current	I_{CC1}	V_{CC} = 5.0 V, V_{IN} = open	01		14	mA
			02		16.2	
			03		8.7	
			04		28	
Supply current	I_{CC2}	V_{CC} = 8.0 V, V_{IN} = open	01		16	mA
			02		18.2	
			03		14	
			04		34	
Collector cutoff current	I_{CEX}	V_{CC} = 5.5 V, V_{IN} = GND, V_{OUT} = 5.5 V	All		50	µA
Propagation delay time low-to-high level from clock to output	t_{PLH1}	V_{CC} = 5.0 V	01,02	35	100	ns
			03		45	
			04	35	75	
Propagation delay time high-to-low level from clock to output	t_{PHL1}	V_{CC} = 5.0 V	01,02	25	90	ns
			03		45	
			04	30	75	
Propagation delay time low-to-high level from set or clear to output	t_{PLH2}	V_{CC} = 5.0 V	01,02	15	125	ns
			03		30	
			04	35	75	
Propagation delay time high-to-low level from set or clear to output	t_{PHL2}	V_{CC} = 5.0 V	01,02	15	100	ns
			03		30	
			04	30	75	

1/ Complete terminal conditions shall be as specified in table III.

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3.7 Manufacturer eligibility. To be eligible to supply microcircuits to this specification, a manufacturer shall have a manufacturer certification in accordance with MIL-M-38510 for at least one line. Not necessarily the line producing the device type described herein.

3.8 Certification. Certification in accordance with MIL-M-38510 is not required for this device.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)
	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1
Final electrical test parameters (method 5004)	1*,2,3,9
Group A test requirements (method 5005)	1,2,3,9
Groups C and D end-point electrical parameters (method 5005)	1,2,3
Additional electrical subgroups for group C periodic inspections	10,11

*PDA applies to subgroup 1 (see 4.2c).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) for class B devices shall be 10 percent based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1, after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection is not required.

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4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510, and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4). Generic test data (see 6.6) may be used to satisfy the requirements for groups C and D inspections. Quality conformance inspection shall be completed on the specific devices covered by this specification before they are shipped.

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, 6, 7, and 8 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical paramters shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical test parameters shall be as specified in table II herein.

b. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.

c. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883 End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging of microcircuits shall be in accordance with MIL-M-38510.

6. NOTES

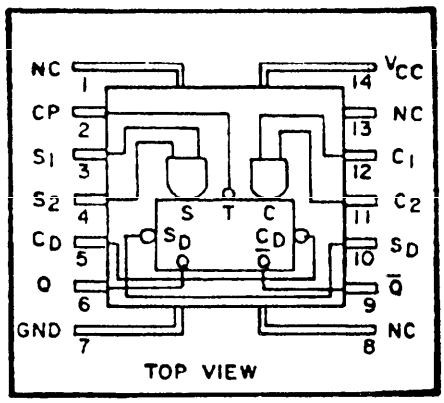
6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for logistic support of existing equipment.

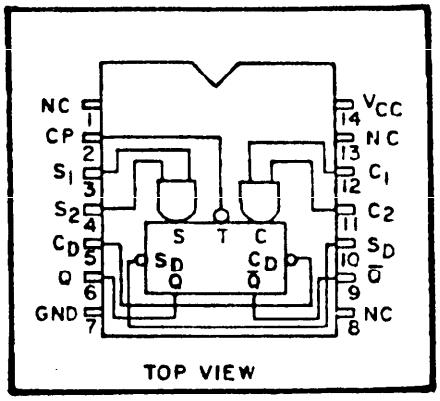
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Device types 01 and 02

Cases A, B, and D

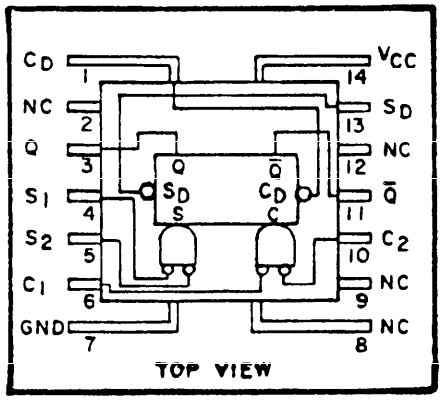


Case C

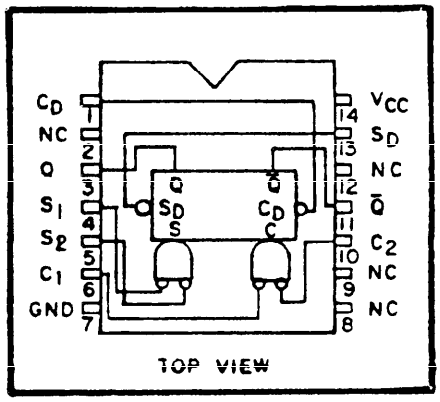


Device type 03

Cases A, B, and D

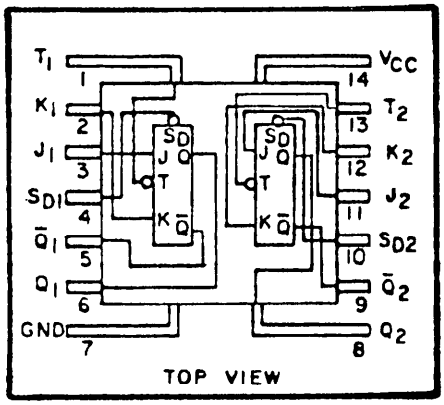


Case C



Device type 04

Cases A, B, and D



Case C

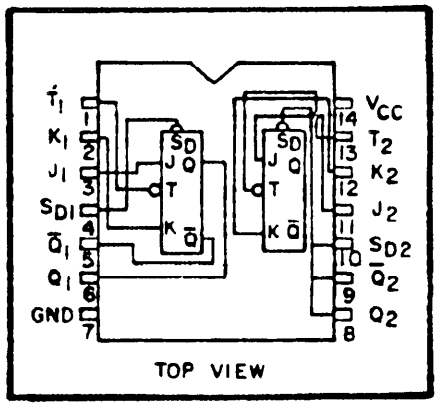


FIGURE 1. Logic diagram and terminal connections.

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Device types 01 and 02

SYNCHRONOUS

INPUTS t_N				OUTPUT t_{N+1}
S_1	S_2	C_1	C_2	Q
L	X	L	X	$\overline{Q}_N$
L	X	X	L	$\overline{Q}_N$
X	L	L	X	$\overline{Q}_N$
X	L	X	L	$\overline{Q}_N$
L	X	H	H	L
X	L	H	H	L
H	H	L	X	H
H	H	X	L	H
H	H	H	H	UNDET.

ASYNCHRONOUS

INPUTS		OUTPUT
S_D	C_D	Q
H	H	$\overline{Q}_N$
H	L	L
L	H	H
L	L	H

J-K MODE

INPUTS t_N		OUTPUT t_{N+1}
S_1	C_1	Q
L	L	$\overline{Q}_N$
L	H	L
H	L	H
H	H	$\overline{Q}_N$

NOTES:
1. t_N = Bit time before clock pulse.
2. t_{N+1} = Bit time after clock pulse.
3. For operation in the J-K mode connect S_2 (pin 4) to Q (pin 6) and C_2 (pin 11) to $\overline{Q}$ (pin 9).

Device type 03

SYNCHRONOUS

INPUTS				OUTPUT
S_2	S_1	C_1	C_2	Q
H	X	H	X	$\overline{Q}_N$
X	H	X	H	$\overline{Q}_N$
L	L	X	H	H
L	L	H	X	H
H	X	L	L	L
X	H	L	L	L
L	L	L	L	UNDET.

ASYNCHRONOUS

INPUTS		OUTPUT
S_D	C_D	Q
H	H	$\overline{Q}_N$
L	H	L
H	L	H
L	L	H

Device type 04

SYNCHRONOUS

INPUTS t_N		OUTPUT t_{N+1}
J	K	Q
L	L	$\overline{Q}_N$
L	H	L
H	L	H
H	H	$\overline{Q}_N$

ASYNCHRONOUS

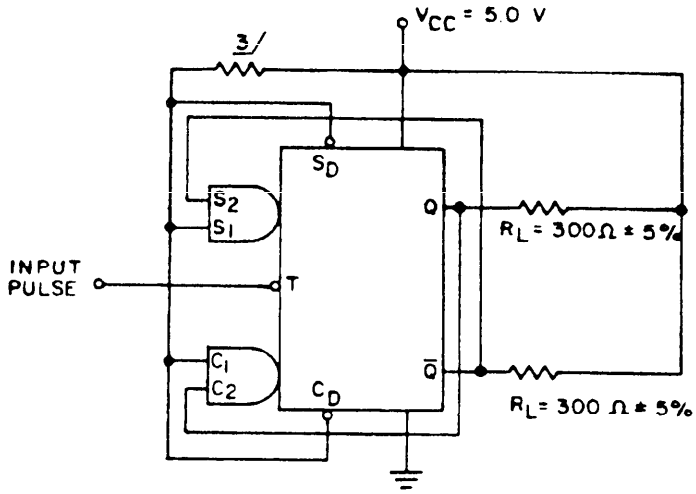
INPUTS	OUTPUTS	
S_D	Q	$\overline{Q}$
L	H	L
H	L	H

NOTES:
1. t_N = Bit time before clock pulse.
2. t_{N+1} = Bit time after clock pulse.

FIGURE 2. Truth tables.

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Device types 01 and 02



Device type 03

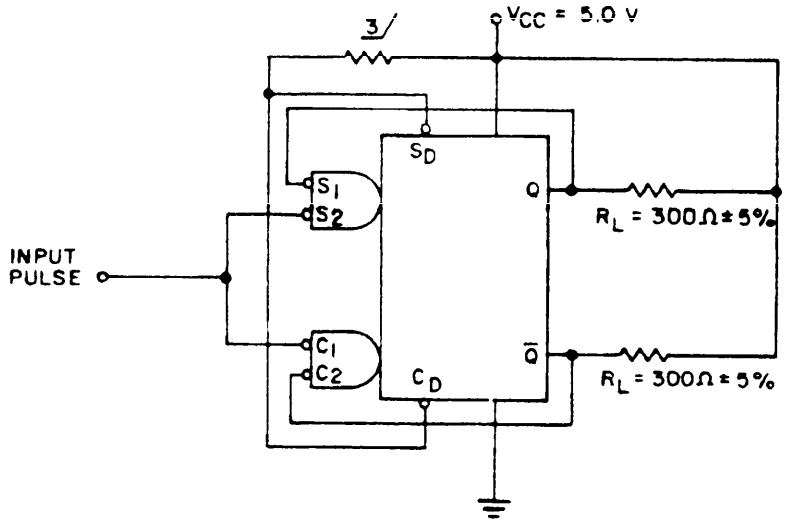
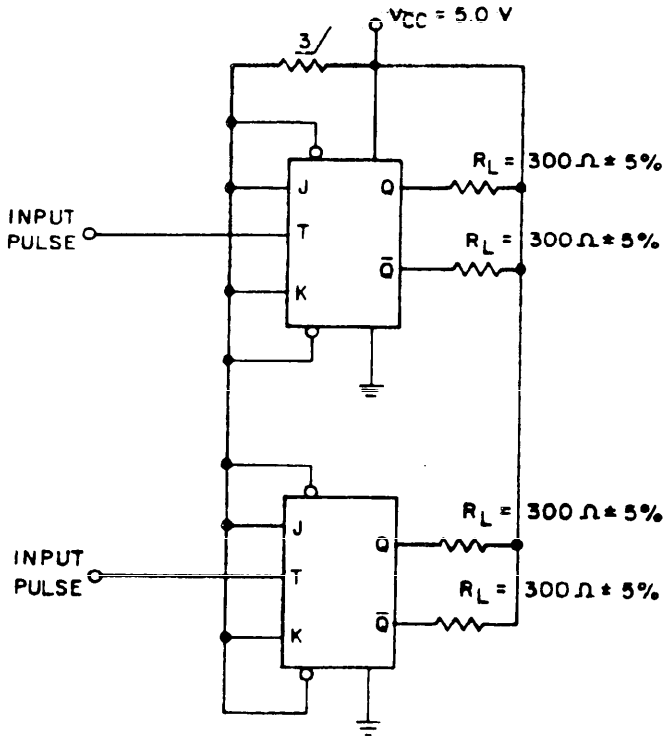


FIGURE 3. Burn-in and life test circuit.

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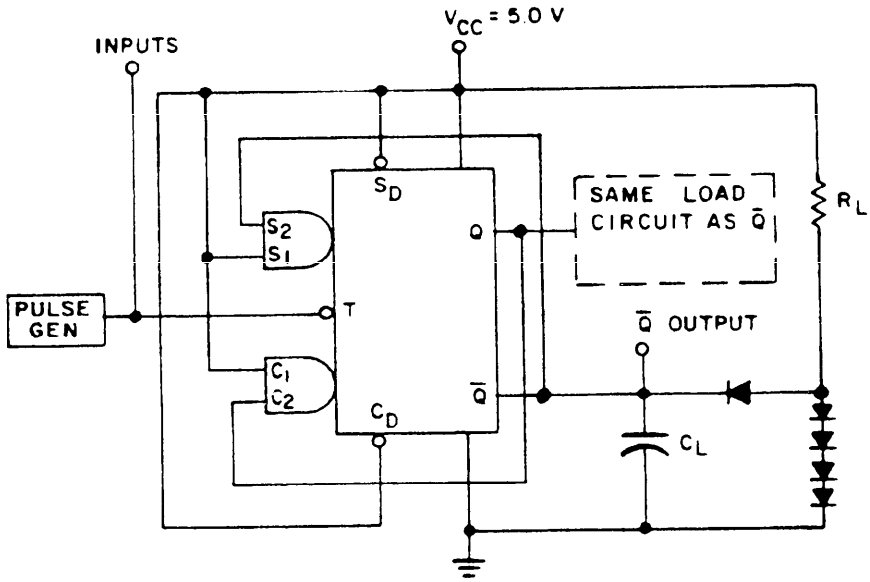
Device type 04



NOTES:
1. The input pulse characteristics are: V_{gen} = 4.0 V, PRR = 100 kHz, duty cycle = 50%, $t_{TLH} = t_{THL}$ = 100 ns.
2. All input adjustments are ±5%.
3. All undriven inputs are connected to V_{CC} thru appropriate current limiting resistors.
4. If parallel excitation: Clock pulse shall be 1 MHz and undriven inputs shall be connected to V_{CC} thru appropriate current limiting resistors.

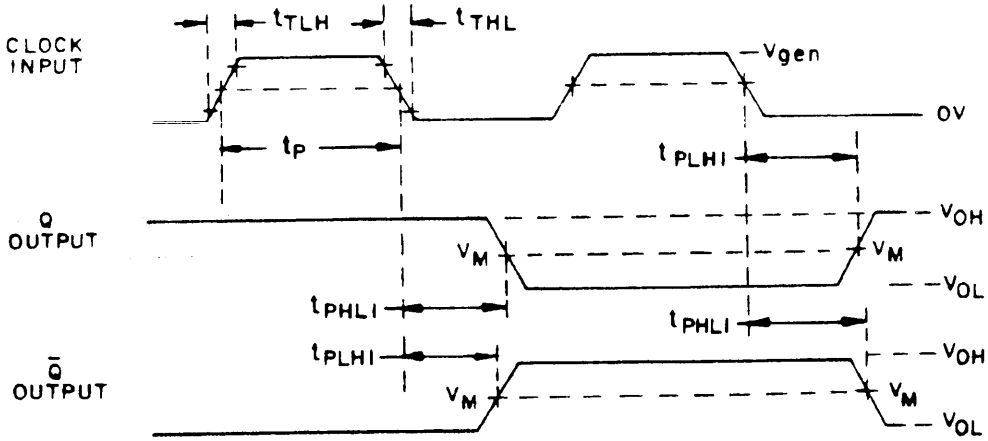
FIGURE 3. Burn-in and life test circuits - Continued.

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	t_{PLH}	t_{PHL}
R_L ±5%	2 kΩ	330Ω
C_L	30 pF	50 pF

TEMP °C	V_M (Volts)
25	1.5
125	1.2
-55	1.7

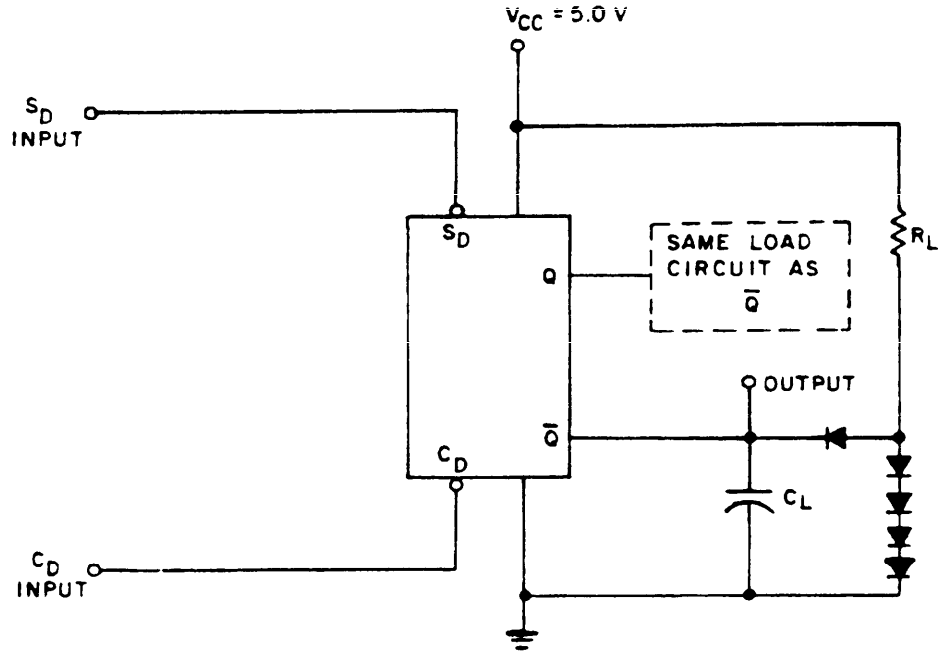


NOTES:
1. The generator has the following characteristics: V_{gen} = 4.0 V, PRR = 1 MHz, t_p = 100 ns, t_{TLH} = t_{THL} = 10 ns.
2. All diodes are 1N4150 or equivalent.
3. All input adjustments are ±5%.

FIGURE 4. Synchronous switching time test circuit for device types 01 and 02.

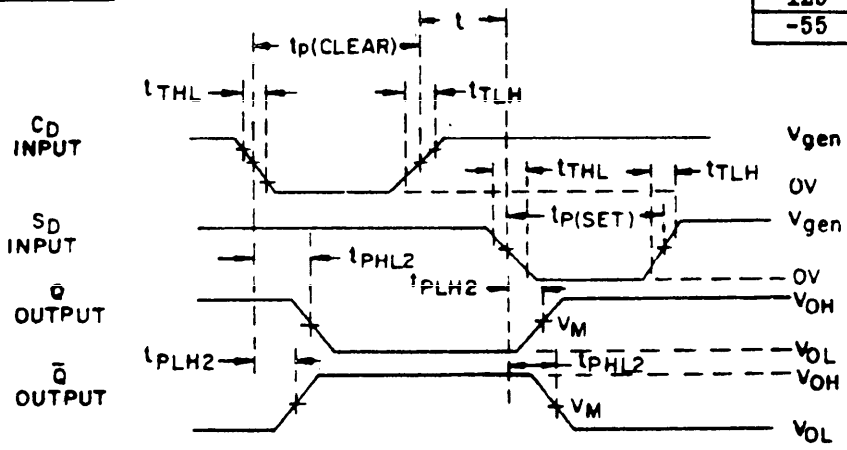
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(3) Only those device types that pass ESDS testing at 1,000 volts or greater shall be



	t_{PLH}	t_{PHL}
R_L ±5%	2 kΩ	330Ω
C_L	30 pF	50 pF

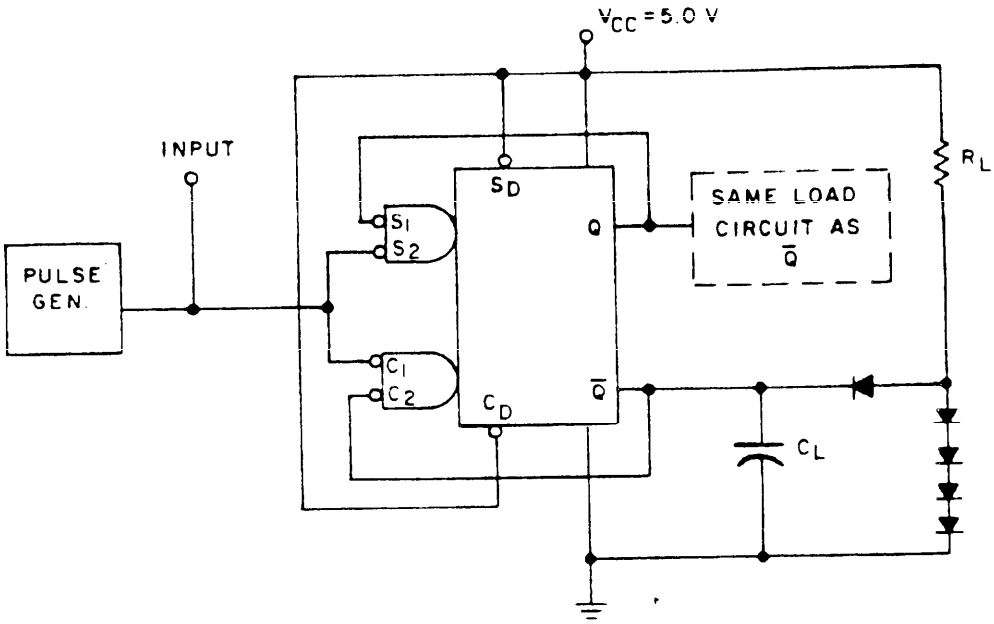
TEMP °C	V_M (Volts)
25	1.5
125	1.2
-55	1.7



NOTES:
1. Direct clear and direct set dominate regardless of the state of clock or inputs.
2. Set and clear direct input pulse characteristics: V_{gen} = 4.0 V, PRR = 250 kHz, t_p = 200 ns, t_{THL} = t_{TLH} = 10 ns, t ≤ 200 ns.
3. All input adjustments are ±5%.
4. All diodes are 1N4150 or equivalent.

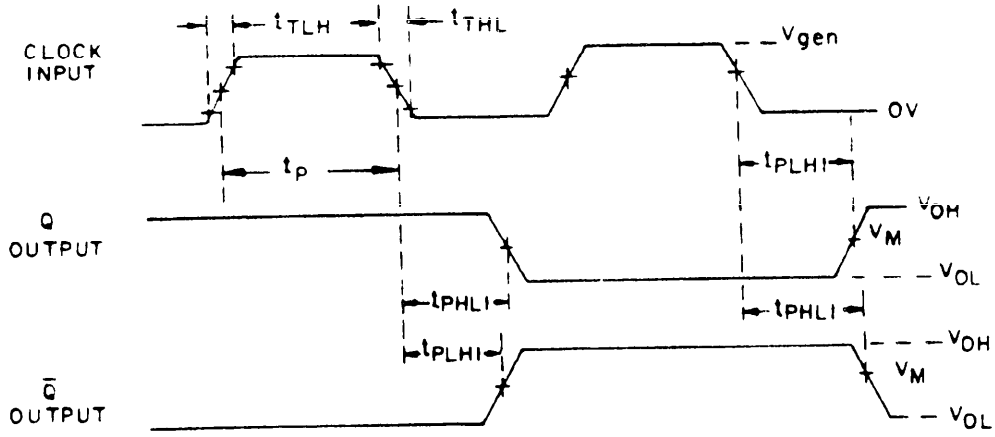
FIGURE 5. Asynchronous switching time test circuit for device types 01 and 02.

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	t_{PLH}	t_{PHL}
R_L ±5%	3.9 kΩ	390Ω
C_L	100 pF	100 pF

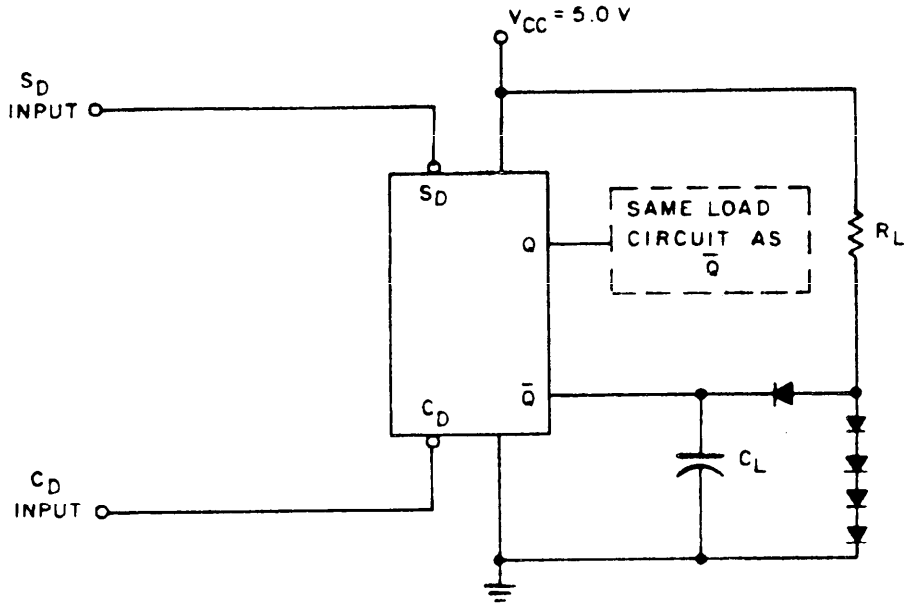
TEMP °C	V_M (Volts)
25	1.5
125	1.2
-55	1.7



NOTES:
1. The generator has the following characteristics: V_{gen} = 4.0 V, PRR = 5 MHz, t_p = 100 ns, t_{TLH} = t_{THL} = 10 nsec.
2. All diodes are 1N4150 or equivalent.
3. All input adjustments are ±5%.

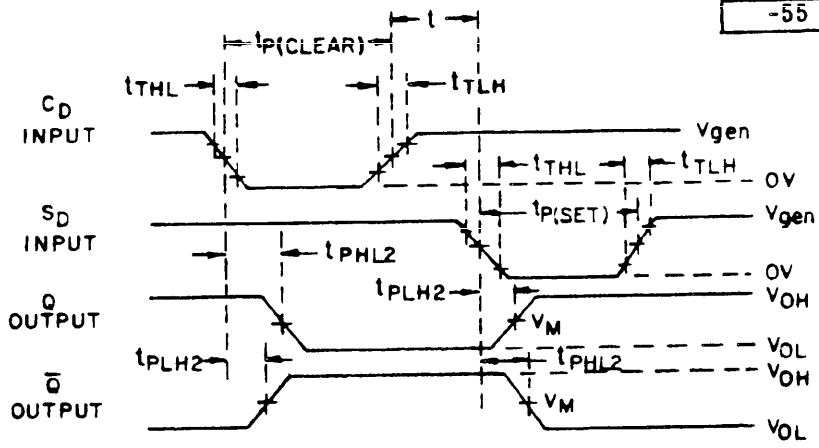
FIGURE 6. Synchronous switching time test circuit for device type 03.

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	t_{PLH}	t_{PHL}
R_L ±5%	3.9 kΩ	390Ω
C_L	100 pF	100 pF

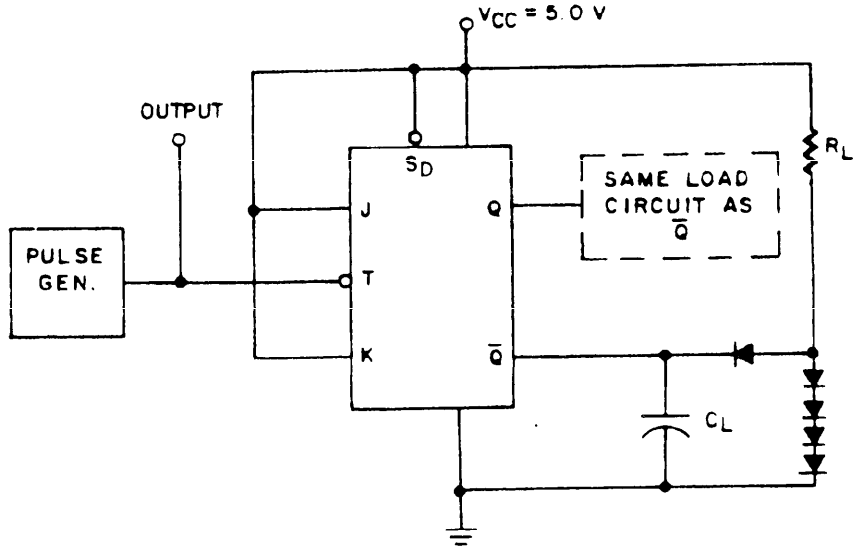
TEMP °C	V_M (Volts)
25	1.5
125	1.2
-55	1.7



NOTES:
1. Direct clear and direct set dominate regardless of the state of clock or inputs.
2. Set and clear direct input pulse characteristics: Vgen = 4.0 V, PRR = 250 kHz, t_p = 200 nsec, t_{THL} = t_{TLH} = 10 ns, t ≤ 200 ns.
3. All input adjustments are ±5%.
4. All diodes are 1N4150 or equivalent.

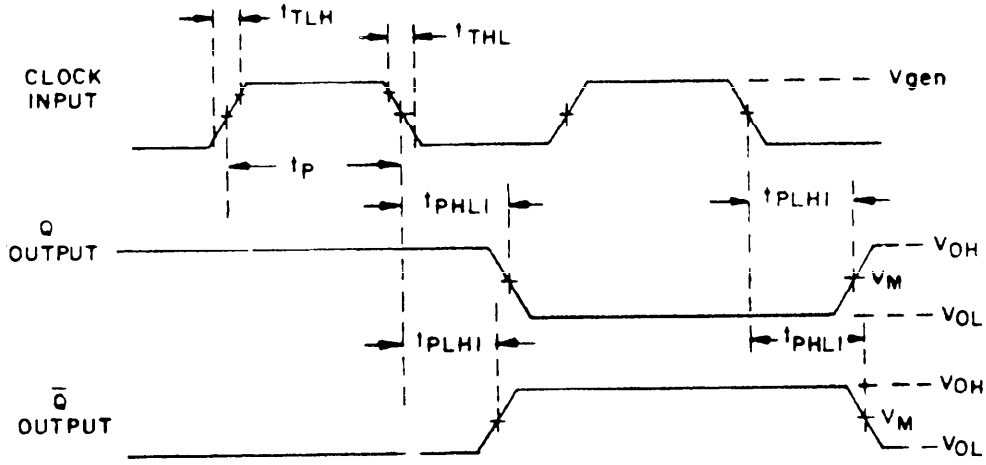
FIGURE 7. Asynchronous switching time test circuit for device type 03.

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	tPLH	tPHL
RL ±5%	2 kΩ	330Ω
CL	30 pF	50 pF

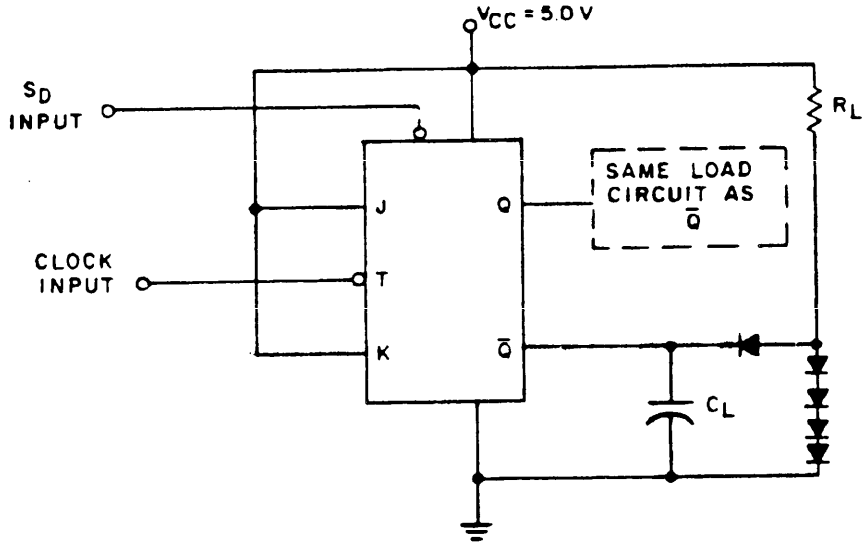
TEMP °C	VM (Volts)
25	1.5
125	1.2
-55	1.7



NOTES:
1. The generator has the following characteristics: Vgen = 4.0 V, PRR = 1 MHz, t_p = 100 ns, t_{TLH} = t_{THL} = 10 ns.
2. All diodes are 1N4150 or equivalent.
3. All input adjustments are ±5%.

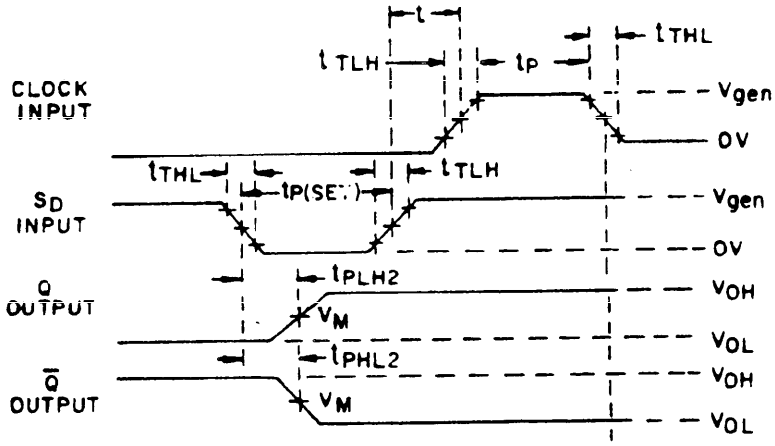
FIGURE 8. Synchronous switching test circuit for device type 04.

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	tPLH	tPHL
R_L ±5%	2 kΩ	330Ω
C_L	30 pF	50 pF

TEMP °C	VM (Volts)
25	1.5
125	1.2
-55	1.7



NOTES:
1. Set direct dominates regardless of the state of clock or J-K inputs.
2. Set direct input pulse characteristics: V_{gen} = 4.0 V, PRR = 1 MHz, t_p = 200 ns, t_{TLH} = t_{THL} = 10 ns, t ≤ 200 ns.
3. All input adjustments are ±5%.
4. All diodes are 1N4150 or equivalent.
5. Clock input pulse characteristics: V_{gen} = 4.0 V, PRR = 1MHz, t_p = 100 ns, t_{TLH} = t_{THL} = 10 ns.

FIGURE 9. Asynchronous switching test circuit for device type 04.

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TABLE III. Group A inspection for device type 01. 1/

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D Case C Test No.	1 NC	2 Clock	3 S_1	4 S_2	5 C_D	6 Q	7 GND	8 NC	9 $\bar{Q}$	10 S_D	11 C_2	12 C_1	13 NC	14 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1	V_{OH}	3006	1		A 2/	1.9 V	1.9 V		-.12 mA	GND				GND	GND		4.5 V	Q	3.4		V
T_C = 25°C			2		A 2/	GND	GND					-0.12mA		1.9 V	1.9 V			$\bar{Q}$			
			3			GND			-0.12mA				1.1 V					Q			
			4					1.1 V				-0.12mA			GND			$\bar{Q}$			
			5		A 2/	GND	GND		-0.12mA				1.1 V	1.1 V	1.1 V			Q			
			6		A 2/	1.1 V	1.1 V	1.1 V				-0.12mA		GND	GND			$\bar{Q}$			
	V_{OL}	3007	7					1.1 V	16 mA				1.9 V					Q		0.4	
	V_{OL}	3007	8					1.9 V				16 mA	1.1 V					$\bar{Q}$		0.4	
	I_{IH1}	3010	9		GND	4.0 V	GND										5.5 V	S_1		2.0	µA
			10			GND	4.0 V											S_2			
			11											4.0 V	GND			C_2			
			12											GND	4.0 V			C_1			
	I_{IH2}		13		A 2/	GND	GND	4.0 V										C_D		4.0	
	I_{IH2}		14		A 2/								4.0 V	GND	GND			S_D		4.0	
	I_{IH3}		15		4.0 V	GND									GND		4.0 V	Clock		10	
	I_{IL1}	3009	16		4.0 V	0.4 V	4.0 V										5.5 V	S_1		-1.07	mA
			17			4.0 V	0.4 V											S_2			
			18											0.4 V	4.0 V			C_2			
			19											4.0 V	0.4 V			C_1			
	I_{IL2}		20			GND		0.4 V					GND		GND			C_D		-3.2	
			21			GND		GND					0.4 V		GND			S_D			
			22		0.4 V			1.1 V										Clock			
			23		0.4 V								1.1 V					Clock			
	I_{OS}	3011	24						GND				GND					$\bar{Q}$	-.7	-1.34	
	I_{OS}	3011	25					GND				GND						Q	-.7	-1.34	
	I_{CEX}		26						5.5 V				GND					$\bar{Q}$		50	µA
	I_{CEX}		27					GND				5.5 V						Q		50	µA
	I_{CC1}	3005	28														5.0 V	V_{CC}		14	mA
	I_{CC2}	3005	29		GND	GND	GND	GND					GND	GND	GND		8.0 V	V_{CC}		16	mA
2	V_{OH}	3006	30		A 2/	1.7 V	1.7 V		-.12 mA					GND	GND		4.5 V	Q	3.3		V
T_C = 125°C			31		A 2/	GND	GND					-.12mA		1.7 V	1.7 V			$\bar{Q}$			
			32			GND			-.12 mA				0.8 V					Q			
			33					0.8 V				-.12mA			GND			$\bar{Q}$			
			34		A 2/	GND	GND		-.12 mA				0.8 V	0.8 V	0.8 V			Q			
			35		A 2/	0.8 V	0.8 V	0.8 V				-.12mA		GND	GND			$\bar{Q}$			
	V_{OL}	3007	36					0.8 V	16 mA				1.7 V					Q		0.45	
	V_{OL}	3007	37					1.7 V				16 mA	0.8 V					$\bar{Q}$		0.45	
	I_{IH1}	3010	38		GND	4.0 V	GND										5.5 V	S_1		5.0	µA
			39			GND	4.0 V											S_2			
			40											4.0 V	GND			C_2			
			41											GND	4.0 V			C_1			
	I_{IH2}		42		A 2/	GND	GND	4.0 V										C_D		10	
	I_{IH2}		43		A 2/								4.0 V	GND	GND			S_D		10	
	I_{IH3}		44		4.0 V	GND									GND		4.0 V	Clock		20	
	I_{IL1}	3009	45		4.0 V	0.4 V	4.0 V										5.5 V	S_1		-1.0	mA
			46			4.0 V	0.4 V											S_2			
			47											0.4 V	4.0 V			C_2			
			48											4.0 V	0.4 V			C_1			

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued. 1/

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C Test No.	1 NC	2 Clock	3 S_1	4 S_2	5 C_D	6 Q	7 GND	8 NC	9 $\bar{Q}$	10 S_D	11 C_2	12 C_1	13 NC	14 V_{CC}	Meas terminal	Test limits Min	Max	Unit
2 T_C = +125°C	I_{IL2}	3009	49			GND		0.4 V		GND			GND		GND		5.5 V	C_D		-2.8	mA
	↓	↓	50			GND		GND					0.4 V		GND			S_D		↓	↓
			51		0.4 V			0.8 V										Clock			
			52		0.4 V								0.8 V					Clock			
	I_{OS}	3011	53						GND				GND					Q	-.615	-1.3	
	I_{OS}	3011	54					GND				GND						$\bar{Q}$	-.615	-1.3	↓
	I_{CEX}		55						5.5 V				GND					Q		50	µA
	I_{CEX}		56					GND				5.5 V					↓	$\bar{Q}$		50	µA
	I_{CC1}	3005	57														5.0 V	V_{CC}		14	mA
	I_{CC2}	3005	58		GND	GND	GND	GND					GND	GND	GND		8.0 V	V_{CC}		16	mA
3 T_C = -55°C	V_{OH}	3006	59		A 2/	2.1 V	2.1 V		-.12 mA					GND	GND		4.5 V	Q	3.3		V
	↓	↓	60		A 2/	GND	GND					-.12mA		2.1 V	2.1 V			$\bar{Q}$	↓		↓
			61			GND			-.12 mA				1.4 V					Q			
			62					1.4 V				-.12mA			GND			$\bar{Q}$			
			63		A 2/	GND	GND		-.12 mA				1.4 V	1.4 V	1.4 V			Q			
			64		A 2/	1.4 V	1.4 V	1.4 V				-.12mA		GND	GND			$\bar{Q}$			
	V_{OL}	3007	65					1.4 V	16 mA				2.1 V					Q		0.4	
	V_{OL}	3007	66					2.1 V				16 mA	1.4 V					$\bar{Q}$		0.4	
	I_{IH1}	3010	67		GND	4.0 V	GND											S_1		2.0	µA
	↓	↓	68			GND	4.0 V											S_2		↓	↓
			69											4.0 V	GND			C_2			
			70											GND	4.0 V			C_1			
	I_{IH2}		71		A 2/	GND	GND	4.0 V										C_D		4.0	
	I_{IH2}		72		A 2/								4.0 V	GND	GND		↓	S_D		4.0	
	I_{IH3}		73		4.0 V	GND									GND		4.0 V	Clock		10	
	I_{IL1}	3009	74		4.0 V	0.4 V	4.0 V										5.5 V	S_1		-1.07	mA
	↓	↓	75			4.0 V	0.4 V											S_2		↓	↓
			76											0.4 V	4.0 V			C_2			
			77											4.0 V	0.4 V			C_1			
	I_{IL2}		78			GND		0.4 V					GND		GND			C_D		-3.2	
	↓	↓	79			GND		GND					0.4 V		GND			S_D		↓	↓
			80		0.4 V			1.4 V										Clock			
			81		0.4 V								1.4 V					Clock			
	I_{OS}	3011	82						GND				GND					Q	-.70	-1.34	
	I_{OS}	3011	83					GND				GND						$\bar{Q}$	-.70	-1.34	↓
	I_{CEX}		84						5.5 V				GND					Q		50	µA
	I_{CEX}		85					GND				5.5 V					↓	$\bar{Q}$		50	µA
	I_{CC1}		86																	14	mA
	I_{CC2}		87		GND	GND	GND	GND					GND	GND	GND		8.0 V	V_{CC}		16	mA
9 T_C = +25°C	t_{PHL1}	3003	88		IN	5.0 V	$\bar{Q}$	5.0 V	OUT				5.0 V	Q	5.0 V		5.0 V	Clock to Q	30	75	ns
	t_{PHL1}	Fig 4	89		↓	↓	↓	↓				OUT		↓	↓			Clock to $\bar{Q}$	30		↓
	t_{PLH1}	↓	90						OUT									Clock to Q	35		
	t_{PLH1}		91									OUT						Clock to $\bar{Q}$	35		
	t_{PHL2}	3003	92					IN	OUT				IN					C_D to Q	15	60	
	t_{PHL2}	Fig 5	93									OUT						S_D to Q		80	
	t_{PLH2}		94									OUT						C_D to $\bar{Q}$		60	
	t_{PLH2}	↓	95						OUT	↓			↓				↓	S_D to $\bar{Q}$	↓	80	↓

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued. 1/

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C Test No.	1 NC	2 Clock	3 S_1	4 S_2	5 C_D	6 Q	7 GND	8 NC	9 $\overline{Q}$	10 S_D	11 C_2	12 C_1	13 NC	14 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
10	t_{PHL1}	3003	96		IN	5.0 V	$\overline{Q}$	5.0 V	OUT	GND			5.0 V	Q	5.0 V		5.0 V	Clock to Q	30	90	
T_C = 125°C	t_{PHL1}	Fig 4	97									OUT						Clock to $\overline{Q}$	30	90	
	t_{PLH1}		98						OUT									Clock to Q	40	100	
	t_{PLH1}		99									OUT						Clock to $\overline{Q}$	40	100	
	t_{PHL2}	3003	100					IN	OUT				IN					C_D to Q	20	100	
	t_{PHL2}	Fig 5	101									OUT						S_D to Q		125	
	t_{PLH2}		102									OUT						C_D to $\overline{Q}$		100	
	t_{PLH2}		103						OUT									S_D to $\overline{Q}$		125	
11	t_{PHL1}	3003	104		IN	5.0 V	$\overline{Q}$	5.0 V	OUT				5.0 V	Q	5.0 V			Clock to Q	25	75	
T_C = -55°C	t_{PHL1}	Fig 4	105									OUT						Clock to $\overline{Q}$	25	75	
	t_{PLH1}		106						OUT									Clock to Q	25	85	
	t_{PLH1}		107									OUT						Clock to $\overline{Q}$	35	85	
	t_{PHL2}	3003	108					IN	OUT				IN					C_D to Q	15	70	
	t_{PHL2}	Fig 5	109									OUT						S_D to Q		80	
	t_{PLH2}		110									OUT						C_D to $\overline{Q}$		70	
	t_{PLH2}		111						OUT									S_D to $\overline{Q}$		80	

1/ Pins not designated may be "high" level logic, "low" level logic, or open.
2/ A = normal clock pulse.

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TABLE III. Group A inspection for device type 02. 1/

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D Case C Test No.	1 NC	2 Clock	3 S_1	4 S_2	5 C_D	6 Q	7 GND	8 NC	9 $\bar{Q}$	10 S_D	11 C_2	12 C_1	13 NC	14 V_{CC}	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
1 T_C = +25°C	V_{OH}	3006	1		A 2/	1.9 V	1.9 V		-.12mA	GND				GND	GND		4.5 V	Q	3.4		V
			2		A 2/	GND	GND					-0.12mA		1.9 V	1.9 V			$\bar{Q}$			
			3			GND			-0.12mA				1.1 V					Q			
			4					1.1 V				-0.12mA			GND			$\bar{Q}$			
			5		A 2/	GND	GND		-0.12mA				1.1 V	1.1 V	1.1 V			Q			
			6		A 2/	1.1 V	1.1 V	1.1 V				-0.12mA		GND	GND			$\bar{Q}$			
	V_{OL}	3007	7					1.1 V	16 mA				1.9 V					Q		0.4	
	V_{OL}	3007	8					1.9 V				16 mA	1.1 V					$\bar{Q}$		0.4	
	I_{IH1}	3010	9		GND	4.0 V	GND										5.5 V	S_1		2.0	µA
			10			GND	4.0 V											S_2			
			11											4.0 V	GND			C_2			
			12											GND	4.0 V			C_1			
	I_{IH2}		13		A 2/	GND	GND	4.0 V										C_D		4.0	
	I_{IH2}		14		A 2/								4.0 V	GND	GND			S_D		4.0	
	I_{IH3}		15		4.0 V	GND									GND		4.0 V	Clock		10	
	I_{IL1}	3009	16		4.0 V	0.4 V	4.0 V										5.5 V	S_1		-1.07	mA
			17			4.0 V	0.4 V											S_2			
			18											0.4 V	4.0 V			C_2			
			19											4.0 V	0.4 V			C_1			
	I_{IL2}		20			GND		0.4 V					GND		GND			C_D		-3.2	
			21			GND		GND					0.4 V		GND			S_D			
			22		0.4 V			1.1 V										Clock			
			23		0.4 V								1.1 V					Clock			
	I_{OS}	3011	24						GND				GND					Q	-2.1	-4.02	
	I_{OS}	3011	25					GND				GND						$\bar{Q}$	-2.1	-4.02	
	I_{CEX}		26						5.5 V				GND					Q		50	µA
	I_{CEX}		27					GND				5.5 V						$\bar{Q}$		50	µA
	I_{CC1}	3005	28														5.0 V	V_{CC}		16.2	mA
	I_{CC2}	3005	29		GND	GND	GND	GND					GND	GND	GND		8.0 V	V_{CC}		18.2	mA
2 T_C = +125°C	V_{OH}	3006	30		A 2/	1.7 V	1.7 V		-.12mA					GND	GND		4.5 V	Q	3.3		V
			31		A 2/	GND	GND					-.12mA		1.7 V	1.7 V			$\bar{Q}$			
			32			GND			-.12mA				0.8 V					Q			
			33					0.8 V				-.12mA			GND			$\bar{Q}$			
			34		A 2/	GND	GND		-.12mA				0.8 V	0.8 V	0.8 V			Q			
			35		A 2/	0.8 V	0.8 V	0.8 V				-.12mA		GND	GND			$\bar{Q}$			
	V_{OL}	3007	36					0.8 V	16 mA				1.7 V					Q		0.45	
	V_{OL}	3007	37					1.7 V				16 mA	0.8 V					$\bar{Q}$		0.45	
	I_{IH1}	3010	38		GND	4.0 V	GND										5.5 V	S_1		5.0	µA
			39			GND	4.0 V											S_2			
			40											4.0 V	GND			C_2			
			41											GND	4.0 V			C_1			
	I_{IH2}		42		A 2/	GND	GND	4.0 V										C_D		10	
	I_{IH2}		43		A 2/								4.0 V	GND	GND			S_D		10	
	I_{IH3}		44		4.0 V	GND									GND		4.0 V	Clock		20	
	I_{IL1}	3009	45		4.0 V	0.4 V	4.0 V										5.5 V	S_1		-1.0	mA
			46			4.0 V	0.4 V											S_2			
			47											0.4 V	4.0 V			C_2			
			48											4.0 V	0.4 V			C_1			

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued. 1/

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D Case C Test No.	1 NC	2 Clock	3 S_1	4 S_2	5 C_D	6 Q	7 GND	8 NC	9 $\overline{Q}$	10 S_D	11 C_2	12 C_1	13 NC	14 V_{CC}	Meas terminal	Test limits Min	Max	Unit
2 T_C +125°C	I_{IL2}	3009	49			GND		0.4 V		GND			GND		GND		5.5 V	C_D		-2.8	mA
			50			GND		GND					0.4 V		GND			S_D			
			51		0.4 V			0.8 V										Clock			
			52		0.4 V								0.8 V					Clock			
	I_{OS}	3011	53						GND				GND					Q	-1.84	-3.9	
	I_{OS}	3011	54					GND				GND						$\overline{Q}$	-1.84	-3.9	
	I_{CEX}		55						5.5 V				GND					Q		50	µA
	I_{CEX}		56					GND				5.5 V						$\overline{Q}$		50	µA
	I_{CC1}	3005	57														5.0 V	V_{CC}		16.2	mA
	I_{CC2}	3005	58		GND	GND	GND	GND					GND	GND	GND		8.0 V	V_{CC}		18.2	mA
3 T_C -55°C	V_{OH}	3006	59		A 2/	2.1 V	2.1 V		-.12 mA					GND	GND		4.5 V	Q	3.3		V
			60		A 2/	GND	GND					-.12mA		2.1 V	2.1 V			$\overline{Q}$			
			61			GND			-.12 mA				1.4 V					Q			
			62					1.4 V				-.12mA			GND			$\overline{Q}$			
			63		A 2/	GND	GND		-.12 mA				1.4 V	1.4 V	1.4 V			Q			
			64		A 2/	1.4 V	1.4 V	1.4 V				-.12mA		GND	GND			$\overline{Q}$			
	V_{OL}	3007	65					1.4 V	16 mA				2.1 V					Q		0.4	
	V_{OL}	3007	66					2.1 V				16 mA	1.4 V					$\overline{Q}$		0.4	
	I_{IH1}	3010	67		GND	4.0 V	GND											S_1		2.0	µA
			68			GND	4.0 V											S_2			
			69											4.0 V	GND			C_2			
			70											GND	4.0 V			C_1			
	I_{IH2}		71		A 2/	GND	GND	4.0 V										C_D		4.0	
	I_{IH2}		72		A 2/								4.0 V	GND	GND			S_D		4.0	
	I_{IH3}		73		4.0 V	GND									GND		4.0 V	Clock		10	
	I_{IL1}	3009	74			0.4 V	4.0 V										5.5 V	S_1		-1.07	mA
			75			4.0 V	0.4 V											S_2			
			76											0.4 V	4.0 V			C_2			
			77											4.0 V	0.4 V			C_1			
	I_{IL2}		78			GND		0.4 V					GND		GND			C_D		-3.2	
			79			GND		GND					0.4 V		GND			S_D			
			80		0.4 V			1.4 V										Clock			
			81		0.4 V								1.4 V					Clock			
	I_{OS}	3011	82						GND				GND					Q	-2.1	-4.02	
	I_{OS}	3011	83					GND				GND						$\overline{Q}$	-2.1	-4.02	
	I_{CEX}		84						5.5 V				GND					Q		50	µA
	I_{CEX}		85					GND				5.5 V						$\overline{Q}$		50	µA
	I_{CC1}		86															V_{CC}		16.2	mA
	I_{CC2}		87		GND	GND	GND	GND					GND	GND	GND			V_{CC}		18.2	mA
9 T_C +25°C	t_{PHL1}	3003	88		IN	5.0 V	$\overline{Q}$	5.0 V	OUT				5.0 V	Q	5.0 V		5.0 V	Clock to Q	30	75	ns
	t_{PHL1}	Fig 4	89									OUT						Clock to $\overline{Q}$	30		
	t_{PLH1}		90						OUT									Clock to Q	35		
	t_{PLH1}		91									OUT						Clock to $\overline{Q}$	35		
	t_{PHL2}	3003	92					IN	OUT				IN					C_D to Q	15	60	
	t_{PHL2}	Fig 5	93									OUT						S_D to $\overline{Q}$		80	
	t_{PLH2}		94									OUT						C_D to $\overline{Q}$		60	
	t_{PLH2}		95						OUT									S_D to Q		80	

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued. 1/

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C Test No.	1 NC	2 Clock	3 S_1	4 S_2	5 C_D	6 Q	7 GND	8 NC	9 $\bar{Q}$	10 S_D	11 C_2	12 C_1	13 NC	14 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
10	t_{PHL1}	3003	96		IN	5.0 V	$\bar{Q}$	5.0 V	OUT	GND			5.0 V	Q	5.0 V		5.0 V	Clock to Q	30	90	ns
T_C = +125°C	t_{PHL1}	Fig 4	97									OUT						Clock to $\bar{Q}$	30	90	
	t_{PLH1}		98						OUT									Clock to Q	40	100	
	t_{PLH1}		99									OUT						Clock to $\bar{Q}$	40	100	
	t_{PHL2}	3003	100					IN	OUT				IN					C_D to Q	20	100	
	t_{PHL2}	Fig 5	101									OUT						S_D to Q		125	
	t_{PLH2}		102									OUT						C_D to $\bar{Q}$		100	
	t_{PLH2}		103						OUT									S_D to $\bar{Q}$		125	
11	t_{PHL1}	3003	104		IN	5.0 V	$\bar{Q}$	5.0 V	OUT				5.0 V	Q	5.0 V			Clock to Q	25	75	
T_C = -55°C	t_{PHL1}	Fig 4	105									OUT						Clock to $\bar{Q}$	25	75	
	t_{PLH1}		106						OUT									Clock to Q	25	85	
	t_{PLH1}		107									OUT						Clock to $\bar{Q}$	35	85	
	t_{PHL2}	3003	108					IN	OUT				IN					C_D to Q	15	70	
	t_{PHL2}	Fig 5	109									OUT						S_D to $\bar{Q}$		80	
	t_{PLH2}		110									OUT						C_D to $\bar{Q}$		70	
	t_{PLH2}		111						OUT									S_D to Q		80	

1/ Pins not designated may be "high" level logic, "low" level logic, or open.
2/ A = normal clock pulse.

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TABLE III. Group A inspection for device type 03. 1/

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D Case C Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas terminal	Test limits Min	Max	Unit
				C_D	NC	Q	S_1	S_2	C_1	GND	NC	NC	C_2	$\bar{Q}$	NC	S_D	V_{CC}				
1	V_{OH}	3006	1	1.9 V		-1.5mA				GND			GND			0.8 V	4.5 V	Q	2.6		V
T_C = +25°C	V_{OH}	3006	2	0.8 V										-1.5mA				$\bar{Q}$	2.6		
	V_{OL}	3007	3	0.8 V		12 mA	GND									1.9 V		Q		0.4	
	V_{OL}	3007	4	1.9 V										12 mA		0.8 V		$\bar{Q}$		0.4	
	I_{IH3}	3010	5				GND	4.0 V					GND				5.5 V	S_2		20	µA
	I_{IH3}		6				GND		4.0 V				GND					C_1		20	
	I_{IH2}		7	4.0 V														C_D		2	
	I_{IH2}		8													4.0 V		S_D		2	
	I_{IL1}	3009	9				0.4 V						4.0 V					S_1		-2.4	mA
	I_{IL1}		10				4.0 V						0.4 V					C_2		-2.4	
	I_{IL2}		11	0.4 V												4.0 V		C_D		-1.82	
	I_{IL2}		12	4.0 V												0.4 V		S_D		-1.82	
	I_{OS}	3011	13			GND										GND		Q	-15.7	-29	
	I_{OS}	3011	14	GND										GND				$\bar{Q}$	-15.7	-29	
	I_{CEX}		15			5.5 V										GND		Q		50	µA
	I_{CEX}		16	GND										5.5 V				$\bar{Q}$		50	µA
	I_{CC1}	3005	17				GND										5.0 V	V_{CC}		8.7	mA
	I_{CC2}	3005	18	GND												GND	8.0 V	V_{CC}		14	mA
2	V_{OH}	3006	19	1.7 V		-1.5mA										0.7 V	4.5 V	Q	2.8		V
T_C = +125°C	V_{OH}	3006	20	0.7 V										-1.5mA		1.7 V		$\bar{Q}$	2.8		
	V_{OL}	3007	21	0.7 V		10.8mA										1.7 V		Q		0.45	
	V_{OL}	3007	22	1.7 V										10.8mA		0.7 V		$\bar{Q}$		0.45	
	I_{IH3}	3010	23				GND	4.0 V					GND				5.5 V	S_2		30	µA
	I_{IH3}		24				GND		4.0 V				GND					C_1		30	
	I_{IH2}		25	4.0 V														C_D		5	
	I_{IH2}		26													4.0 V		S_D		5	
	I_{IL1}	3009	27				0.4 V						4.0 V					S_1		-2.25	mA
	I_{IL1}		28				4.0 V						0.4 V					C_2		-2.25	
	I_{IL2}		29	0.4 V												4.0 V		C_D		-1.62	
	I_{IL2}		30	4.0 V												0.4 V		S_D		-1.62	
	I_{OS}	3011	31			GND										GND		Q	-14	-29	
	I_{OS}	3011	32	GND										GND				$\bar{Q}$	-14	-29	
	I_{CEX}		33			5.5 V										GND		Q		50	µA
	I_{CEX}		34	GND										5.5 V				$\bar{Q}$		50	µA
	I_{CC1}	3005	35				GND										5.0 V	V_{CC}		8.7	mA
	I_{CC2}	3005	36	GND												GND	8.0 V	V_{CC}		14	mA
3	V_{OH}	3006	37	2.1 V		-1.5mA							GND			0.95 V	4.5 V	Q	2.5		V
T_C = -55°C	V_{OH}	3006	38	0.95 V										-1.5mA		2.1 V		$\bar{Q}$	2.5		
	V_{OL}	3007	39	0.95 V		11.4mA										0.95 V		Q		0.4	
	V_{OL}	3007	40	2.1 V										11.4mA		1.4 V		$\bar{Q}$		0.4	
	I_{IH3}	3010	41				GND	4.0 V					GND				5.5 V	S_2		20	µA
	I_{IH3}		42				GND		4.0 V				GND					C_1		20	
	I_{IH2}		43	4.0 V														C_D		2	
	I_{IH2}		44													4.0 V		S_D		2	

See footnote at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued. 1/

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits Min	Max	Unit
				C_D	NC	Q	S_1	S_2	C_1	GND	NC	NC	C_2	$\bar{Q}$	NC	S_D	V_{CC}				
3 T_C = -55°C	I_{IL1}	3009	45				0.4 V			GND			4.0 V				5.5 V	S_1		-2.4	mA
	I_{IL1}	↓	46				4.0 V			↓			0.4 V				↓	C_2		-2.4	
	I_{IL2}		47	0.4 V												4.0 V		C_D		-1.82	
	I_{IL2}	↓	48	4.0 V												0.4 V		S_D		-1.82	
	I_{OS}	3011	49			GND										GND		Q or $\bar{Q}$	-15.7	-29	
	I_{OS}	3011	50	GND										GND				Q or $\bar{Q}$	-15.7	-29	↓
	I_{CEX}		51			5.5 V										GND		Q or $\bar{Q}$		50	µA
	I_{CEX}		52	GND										5.5 V				Q or $\bar{Q}$		50	µA
	I_{CC1}	3005	53				GND										5.0 V	V_{CC}		8.0	mA
	I_{CC2}	3005	54	GND													8.0 V	V_{CC}		14	mA
9 T_C = +25°C	t_{PHL1}	3003	55	5.0 V			Q	IN	S_2				$\bar{Q}$			5.0 V	5.0 V	S_2 to Q		30	ns
	t_{PHL1}	Fig 6	56	↓			↓	↓	↓				↓			↓	↓	S_2 to $\bar{Q}$		↓	↓
	t_{PLH1}		57															S_2 to Q			
	t_{PLH1}	↓	58	IN												IN		S_2 to $\bar{Q}$			
	t_{PHL2}	3003	59			OUT								OUT				C_D to Q			
	t_{PHL2}	Fig 7	60											OUT				S_D to $\bar{Q}$			
	t_{PLH2}		61															C_D to $\bar{Q}$			
	t_{PLH2}	↓	62	↓		OUT	↓	↓	↓				↓			↓	↓	S_D to Q		↓	↓
10 T_C = +125°C	t_{PHL1}	3003	63	5.0 V			Q	IN	S_2				$\bar{Q}$			5.0 V		S_2 to Q		45	ns
	t_{PHL1}	Fig 6	64	↓			↓	↓	↓				↓			↓		S_2 to $\bar{Q}$		↓	↓
	t_{PLH1}		65															S_2 to Q			
	t_{PLH1}	↓	66	IN												IN		S_2 to $\bar{Q}$			
	t_{PHL2}	3003	67			OUT								OUT				C_D to Q			
	t_{PHL2}	Fig 7	68											OUT				S_D to $\bar{Q}$			
	t_{PLH2}		69															C_D to $\bar{Q}$			
	t_{PLH2}	↓	70	↓		OUT	↓	↓	↓				↓			↓	↓	S_D to Q		↓	↓
11 T_C = -55°C	t_{PHL1}	3003	71	5.0 V			Q	IN	S_2				$\bar{Q}$			5.0 V		S_2 to Q		45	ns
	t_{PHL1}	Fig 6	72	↓			↓	↓	↓				↓			↓		S_2 to $\bar{Q}$		↓	↓
	t_{PLH1}		73															S_2 to Q			
	t_{PLH1}	↓	74	IN												IN		S_2 to $\bar{Q}$			
	t_{PHL2}	3003	75			OUT								OUT				C_D to Q			
	t_{PHL2}	Fig 7	76											OUT				S_D to $\bar{Q}$			
	t_{PLH2}		77															C_D to $\bar{Q}$			
	t_{PLH2}	↓	78	↓		OUT	↓	↓	↓	↓			↓			↓	↓	S_D to Q		↓	↓

1/ Pins not designated may be "high" level logic, "low" level logic, or open.

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TABLE III. Group A inspection for device type 04. 1/

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits Min	Max	Unit
				T_1	K_1	J_1	S_{D1}	$\bar{Q}_1$	Q_1	GND	Q_2	$\bar{Q}_2$	S_{D2}	J_2	K_2	T_2	V_{CC}				
1	V_{OH}	3006	1	A 2/	GND	1.9 V	4.5 V		-.12 mA	GND							4.5 V	Q_1	3.4		V
T_C = +25°C			2	A 2/	1.9 V	GND	4.5 V	-.12mA										$\bar{Q}_1$			
			3		1.9 V	GND	1.1 V		-.12 mA									Q_1			
			4								-.12mA		4.5 V	1.9 V	GND	A 2/		$\bar{Q}_2$			
			5									-.12mA	4.5 V	GND	1.9 V	A 2/		Q_2			
			6								-.12mA		1.1 V	GND	1.9 V			$\bar{Q}_2$			
	V_{OL}	3007	7	A 2/	1.9 V	GND	4.5 V		-15.2mA									Q_1		0.4	
			8	A 2/	GND	1.9 V	4.5 V	-15.2mA										$\bar{Q}_1$			
			9		1.9 V	GND	1.1 V	-15.2mA										Q_1			
			10								-15.2mA		4.5 V	GND	1.9 V	A 2/		$\bar{Q}_2$			
			11									-15.2mA	4.5 V	1.9 V	GND	A 2/		$\bar{Q}_2$			
			12									-15.2mA	1.1 V	GND	1.9 V			Q_2			
	I_{IH1}	3010	13	GND	GND	4.0 V											5.5 V	J_1		2	µA
			14	GND	4.0 V	GND												K_1			
			15											4.0 V	GND	GND		J_2			
			16											GND	4.0 V	GND		K_2			
	I_{IH2}		17	A 2/	GND	GND	4.0 V											S_{D1}		4	
	I_{IH2}		18										4.0 V	GND	GND	A 2/		S_{D2}		4	
	I_{IH3}		19	4.0 V	GND	GND	GND										4.0 V	T_1		10	
	I_{IH3}		20										GND	GND	GND	4.0 V	4.0 V	T_2		10	
	I_{IL1}	3009	21	4.0 V	4.0 V	0.4 V											5.5 V	J_1		-0.08	mA
			22	4.0 V	0.4 V		0.4 V											K_1			
			23											0.4 V	4.0 V	4.0 V		J_2			
			24										0.4 V		0.4 V	4.0 V		K_2			
	I_{IL2}		25			0.4 V	0.4 V											S_{D1}		-2.2	
			26										0.4 V	0.4 V				S_{D2}		-2.2	
			27	0.4 V			0.4 V											T_1		-2.93	
			28										0.4 V		0.4 V			T_2		-2.93	
	I_{OS}	3011	29				GND		GND									Q_1	-.7	-1.33	
			30	A 2/	4.0 V	GND	4.0 V	GND										$\bar{Q}_1$			
			31								GND		GND					Q_2			
			32									GND	4.0 V	GND	4.0 V	A 2/		$\bar{Q}_2$			
	I_{CEX}		33				GND		5.5 V									Q_1		50	µA
			34	A 2/	4.0 V	GND	4.0 V	5.5 V										$\bar{Q}_1$			
			35								5.5 V		GND					Q_2			
			36									5.5 V	4.0 V	GND	4.0 V	A 2/		$\bar{Q}_2$			
	I_{CC1}	3005	37				GND						GND				5.0 V	V_{CC}		28	mA
	I_{CC2}	3005	38	GND	GND	GND	GND						GND	GND	GND	GND	8.0 V	V_{CC}		34	mA
2	V_{OH}	3006	39	A 2/	GND	1.7 V	4.5 V		-12 mA								4.5 V	Q_1	3.4		V
T_C = +125°C			40	A 2/	1.7 V	GND	4.5 V	-12 mA										$\bar{Q}_1$			
			41		1.7 V	GND	0.8 V		-12 mA									Q_1			
			42								-12 mA		4.5 V	1.7 V	GND	A 2/		$\bar{Q}_2$			
			43									-12 mA	4.5 V	GND	1.7 V	A 2/		Q_2			
			44								-12 mA		0.8 V	GND	1.7 V			$\bar{Q}_2$			
	V_{OL}	3007	45	A 2/	1.7 V	GND	4.5 V		14.6mA									Q_1		0.45	
			46	A 2/	GND	1.7 V	4.5 V	14.6mA										$\bar{Q}_1$			
			47		1.7 V	GND	0.8 V	14.6mA										Q_1			
			48								14.6mA		4.5 V	GND	1.7 V	A 2/		$\bar{Q}_2$			
			49									14.6mA	4.5 V	1.7 V	GND	A 2/		Q_2			
			50									14.6mA	0.8 V	GND	1.7 V			$\bar{Q}_2$			

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued. 1/

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
				T_1	K_1	J_1	S_{D1}	$\overline{Q}_1$	Q_1	GND	Q_2	$\overline{Q}_2$	S_{D2}	J_2	K_2	T_2	V_{CC}				
2	I_{IH1}	3010	51	GND	GND	4.0 V				GND							5.5 V	J_1		5	µA
T_C = +125°C			52	GND	4.0 V	GND												K_1			
			53											4.0 V	GND	GND		J_2			
			54											GND	4.0 V	GND		K_2			
	I_{IH2}		55	A 2/	GND	GND	4.0 V											S_{D1}		10	
	I_{IH2}		56										4.0 V	GND	GND	A 2/		S_{D2}		10	
	I_{IH3}		57	4.0 V	GND	GND	GND										4.0 V	T_1		20	
			58										GND	GND	GND	4.0 V	4.0 V	T_2		20	
	I_{IL1}	3009	59	4.0 V	4.0 V	0.4 V											5.5 V	J_1		-0.92	mA
			60	4.0 V	0.4 V		0.4 V											K_1			
			61											0.4 V	4.0 V	4.0 V		J_2			
			62										0.4 V		0.4 V	4.0 V		K_2			
	I_{IL2}		63			0.4 V	0.4 V											S_{D1}		-2.2	
			64										0.4 V	0.4 V				S_{D2}		-2.2	
			65	0.4 V			0.4 V											T_1		-2.57	
			66										0.4 V			0.4 V		T_2		-2.57	
	I_{OS}	3011	67				GND		GND									Q_1	-.62	-1.3	
			68	A 2/	4.0 V	GND	4.0 V	GND										$\overline{Q}_1$			
			69								GND		GND					Q_2			
			70									GND	4.0 V	GND	4.0 V	A 2/		$\overline{Q}_2$			
	I_{CEX}		71				GND		5.5 V									Q_1		50	µA
			72	A 2/	4.0 V	GND	4.0 V	5.5 V										$\overline{Q}_1$			
			73								5.5 V		GND					Q_2			
			74									5.5 V	4.0 V	GND	4.0 V	A 3/		$\overline{Q}_2$			
	I_{CC1}	3005	75				GND						GND				5.0 V	V_{CC}		28	mA
	I_{CC2}	3005	76	GND	GND	GND	GND						GND	GND	GND	GND	8.0 V	V_{CC}		34	mA
3	V_{OH}	3006	77	A 2/	GND	2.1 V	4.5 V		-.12mA								4.5 V	Q_1	3.4		V
T_C = -55°C			78	A 2/	2.1 V	GND	4.5 V	-.12mA										$\overline{Q}_1$			
			79		2.1 V	GND	1.4 V		-.12mA									Q_1			
			80								-.12mA		4.5 V	2.1 V	GND	A 2/		$\overline{Q}_2$			
			81									-.12mA	4.5 V	GND	2.1 V	A 2/		Q_2			
			82								-.12mA		1.4 V	GND	2.1 V			$\overline{Q}_2$			
	V_{OL}	3007	83	A 2/	2.1 V	GND			-14.6mA									Q_1		0.4	
			84	A 2/	GND	2.1 V		-14.6mA										$\overline{Q}_1$			
			85		2.1 V	GND	1.4 V	-14.6mA										Q_2			
			86								-14.6mA			GND	2.1 V	A 2/		$\overline{Q}_2$			
			87									-14.6mA		2.1 V	GND	A 2/		Q_2			
			88									-14.6mA	1.4 V	GND	2.1 V			$\overline{Q}_2$			
	I_{IH1}	3010	89	GND	GND	4.0 V											5.5 V	J_1		2	µA
			90	GND	4.0 V	GND												K_1			
			91											4.0 V	GND	GND		J_2			
			92											GND	4.0 V	GND		K_2			
	I_{IH2}		93	A 2/	GND	GND	4.0 V											S_{D1}		4	
	I_{IH2}		94										4.0 V	GND	GND	A 2/		S_{D2}		4	
	I_{IH3}		95	4.0 V	GND	GND	GND										4.0 V	T_1		10	
	I_{IH3}		96										GND	GND	GND	4.0 V	4.0 V	T_2		10	

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued. 1/

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits Min	Max	Unit
				T_1	K_1	J_1	S_{D1}	$\bar{Q}_1$	Q_1	GND	Q_2	$\bar{Q}_2$	S_{D2}	J_2	K_2	T_2	V_{CC}				
3 T_C = -55°C	I_{IL1}	3009	97	4.0 V	4.0 V	4.0 V				GND							5.5 V	J_1		-0.98	mA
			98	4.0 V	0.4 V		0.4 V											K_1			
			99											0.4 V	4.0 V	4.0 V		J_2			
			100										0.4 V		0.4 V	4.0 V		K_2			
	I_{IL2}		101			0.4 V	0.4 V											S_{D1}		-2.2	
			102										0.4 V	0.4 V				S_{D2}		-2.2	
			103	0.4 V			0.4 V											T_1		-2.9	
			104										0.4 V		0.4 V			T_2		-2.9	
	I_{OS}	3011	105				GND		GND									Q_1	-.7	-1.33	
			106	A 2/	4.0 V	GND	4.0 V	GND										$\bar{Q}_1$			
			107								GND		GND					Q_2			
			108									GND	4.0 V	GND	4.0 V	A 2/		$\bar{Q}_2$			
	I_{CEX}		109				GND		5.5 V									Q_1		50	µA
			110	A 2/	4.0 V	GND	4.0 V	5.5 V										$\bar{Q}_1$			
			111								5.5 V		GND					Q_2			
			112									5.5 V	4.0 V	GND	4.0 V	A 2/		$\bar{Q}_2$			
	I_{CC1}	3005	113				GND						GND				5.0 V	V_{CC}		28	mA
	I_{CC2}	3005	114	GND	GND	GND	GND						GND	GND	GND	GND	8.0 V	V_{CC}		34	mA
9 T_C = +25°C	t_{PHL1}	3003	115	IN	5.0 V	5.0 V	5.0 V		OUT								5.0 V	T_1 to Q_1	30	75	ns
	t_{PHL1}	Fig 8	116					OUT										T_1 to $\bar{Q}_1$	30		
	t_{PLH1}		117						OUT									T_1 to Q_1	35		
	t_{PLH1}		118					OUT										T_1 to $\bar{Q}_1$	35		
	t_{PHL2}	3003	119				IN		OUT									S_{D1} to Q_1	30		
	t_{PHL2}	Fig 9	120					OUT										S_{D1} to $\bar{Q}_1$	30		
	t_{PLH2}		121						OUT									S_{D1} to Q_1	35		
	t_{PLH2}		122					OUT										S_{D1} to $\bar{Q}_1$	35		
	t_{PHL1}	3003	123								OUT		5.0 V	5.0 V	5.0 V	IN		T_2 to Q_2	30		
	t_{PHL1}	Fig 8	124									OUT						T_2 to $\bar{Q}_2$	30		
	t_{PLH1}		125								OUT							T_2 to Q_2	35		
	t_{PLH1}		126									OUT						T_2 to $\bar{Q}_2$	35		
	t_{PHL2}	3003	127								OUT		IN					S_{D2} to Q_2	30		
	t_{PHL2}	Fig 9	128									OUT						S_{D2} to $\bar{Q}_2$	30		
	t_{PLH2}		129								OUT							S_{D2} to Q_2	35		
	t_{PLH2}		130									OUT						S_{D2} to $\bar{Q}_2$	35		

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued. 1/

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C Test No.	1 T_1	2 K_1	3 J_1	4 S_{D1}	5 $\overline{Q}_1$	6 Q_1	7 GND	8 Q_2	9 $\overline{Q}_2$	10 S_{D2}	11 J_2	12 K_2	13 T_2	14 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
10	t_{PHL1}	3003	131	IN	5.0 V	5.0 V	5.0 V		OUT	GND							5.0 V	T_1 to Q_1	30	90	ns
T_C = +125°C	t_{PHL1}	Fig 8	132					OUT										T_1 to $\overline{Q}_1$	30		
	t_{PLH1}		133						OUT									T_1 to Q_1	35		
	t_{PLH1}		134					OUT										T_1 to $\overline{Q}_1$	35		
	t_{PHL2}	3003	135				IN		OUT									S_{D1} to Q_1	30		
	t_{PHL2}	Fig 9	136					OUT										S_{D1} to $\overline{Q}_1$	30		
	t_{PLH2}		137						OUT									S_{D1} to Q_1	35		
	t_{PLH2}		138					OUT										S_{D1} to $\overline{Q}_1$	35		
	t_{PHL1}	3003	139								OUT		5.0 V	5.0 V	5.0 V	IN		T_2 to Q_2	30		
	t_{PHL1}	Fig 8	140									OUT						T_2 to $\overline{Q}_2$	30		
	t_{PLH1}		141								OUT							T_2 to Q_2	35		
	t_{PLH1}		142									OUT						T_2 to $\overline{Q}_2$	35		
	t_{PHL2}	3003	143								OUT		IN					S_{D2} to Q_2	30		
	t_{PHL2}	Fig 9	144									OUT						S_{D2} to $\overline{Q}_2$	30		
	t_{PLH2}		145								OUT							S_{D2} to Q_2	35		
	t_{PLH2}		146									OUT						S_{D2} to $\overline{Q}_2$	35		
11	t_{PHL1}	3003	147	IN	5.0 V	5.0 V	5.0 V		OUT									T_1 to Q_1	30		
T_C = -55°C	t_{PHL1}	Fig 8	148					OUT										T_1 to $\overline{Q}_1$	30		
	t_{PLH1}		149						OUT									T_1 to Q_1	35		
	t_{PLH1}		150					OUT										T_1 to $\overline{Q}_1$	35		
	t_{PHL2}	3003	151				IN		OUT									S_{D1} to Q_1	30		
	t_{PHL2}	Fig 9	152					OUT										S_{D1} to $\overline{Q}_1$	30		
	t_{PLH2}		153						OUT									S_{D1} to Q_1	35		
	t_{PLH2}		154					OUT										S_{D1} to $\overline{Q}_1$	35		
	t_{PHL1}	3003	155								OUT		5.0 V	5.0 V	5.0 V	IN		T_2 to Q_2	30		
	t_{PHL1}	Fig 8	156									OUT						T_2 to $\overline{Q}_2$	30		
	t_{PLH1}		157								OUT							T_2 to Q_2	35		
	t_{PLH1}		158									OUT						T_2 to $\overline{Q}_2$	35		
	t_{PHL2}	3003	159								OUT		IN					S_{D2} to Q_2	30		
	t_{PHL2}	Fig 9	160									OUT						S_{D2} to $\overline{Q}_2$	30		
	t_{PLH2}		161								OUT							S_{D2} to Q_2	35		
	t_{PLH2}		162									OUT						S_{D2} to $\overline{Q}_2$	35		

1/ Pins not designated may be "high" level logic, "low" level logic, or open.
2/ A = normal clock pulse.

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6.3 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity, if applicable.

e. Requirements for packaging and packing.

f. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Electrical ground (common terminal)
I_{CEX}	Collector cutoff current
I_{IN}	Current flowing into an input terminal
V_{IN}	Voltage level at an input terminal

6.5 Logistic support. Lead materials and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Generic test data. Generic test data may be used to satisfy the requirements of 4.4.3. Group C generic test data shall be on date codes no more than one year old and on a die in the same microcircuit group (see appendix E of MIL-M-38510) with the same material, design and process and from the same plant as the die represented. Group D (see 4.4.4) generic data shall be on date codes no more than one year old and on the same package type (see terms, definitions, and symbols of MIL-M-38510) and from the same plant as the package represented. The vendor is required to retain the generic data for a period of not less than 36 months from the date of shipment.

6.7 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	945
02	948
03	950
04	9093

6.8 Ordering guidance. Since the qualification and certification requirements have been removed from the specification, orders may be placed immediately.

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Navy - EC
Air Force - 17

Review activities:
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Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/33A	MICROCIRCUITS, DIGITAL, DTL, FLIP FLOPS, MONOLITHIC SILICON

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

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